Two-Stage Retrieval for RAG: Bi-Encoder Baseline vs. Cross-Encoder Re-Ranking

Vishnuraj K A
2025900017

1 Method

1.1 Document and Preprocessing

I used Apple Inc.'s FY2025 Form 10-K (80 pages), a regulatory filing that mixes narrative risk disclosures, legal proceedings, financial tables and dense legal exhibits on debt securities. This document contains exceptions (shall not apply, except), constraints (in whole, but not in part) and negations, which makes it a good fit for this retrieval task.

Text is extracted page by page with PyMuPDF. Repeated page footers are detected as lines that occur on $\geq$30% of pages and are removed. Pages are split into 1000-character chunks with 150-character overlap, splitting at sentence or word boundaries, which gives 376 chunks.

1.2 Retrieval Pipelines

Both pipelines share the same chunk corpus, index and top-10 candidate list (Table 1).

Pipeline 1 (Baseline). Chunks are embedded once with a bi-encoder and stored in a FAISS flat inner-product index. With L2-normalised vectors, inner product equals cosine similarity. For a query, the top-10 chunks are retrieved and the top-3 are passed to the LLM.

Pipeline 2 (Re-Ranking). The same top-10 candidates are scored as (query, chunk) pairs by a cross-encoder, re-sorted by score, and the new top-3 are passed to the LLM.

Table 1: Experimental setup.

Bi-encoder	`BAAI/bge-m3` (1024-d), FAISS `IndexFlatIP`
Cross-encoder	`cross-encoder/ms-marco-MiniLM-L-6-v2`
LLM	`Qwen/Qwen3-8B`
Retrieval	$k = 10$ candidates, $n = 3$ chunks to the LLM

The two encoders differ in where query–document processing happens (Table 2). The bi-encoder compresses query and chunk independently into single vectors, so relevance is reduced to how close both vectors are in the embedding space. The cross-encoder processes both texts together, letting every query token attend to every chunk token before producing one relevance score.

Table 2: Bi-encoder vs. cross-encoder.

	Bi-Encoder	**Cross-Encoder**
Input	Query and chunk encoded separately	(query, chunk) encoded as one sequence
Interaction	None until a final dot product	Full self-attention across query and chunk tokens
Output	Embedding vector	One relevance score
Pre-compute	Chunk vectors indexed once offline	None; every pair scored at query time
Cost per query	1 encoder pass + index search	N encoder passes
Weakness	Topic similarity; blurs negation and exceptions	Too slow for the full corpus

1.3 Generation

Both pipelines use the same prompt template and decoding settings. The system prompt instructs the LLM to answer only from the context, state any conditions or exceptions, and reply "The context does not contain this information" otherwise. Only the three chunks differ between Answer 1 and Answer 2.

1.4 Evaluation Protocol

The queries were written for each failure category by looking at the corpus and keeping those where the gold chunk (the chunk that actually answers the query) sat at ranks 4–10 for the baseline. Gold chunks are identified by a unique phrase from their text, so the labels survive re-chunking. Answers were judged correct if they state the fact in the gold chunk without contradicting it. Latency was averaged over 10 queries.

2 Results

2.1 Rank Shifts

Table 3 lists the seven failure queries. In every case the gold chunk was outside the baseline top-3 and was promoted into the top-3 by the cross-encoder, four times to rank 1. The baseline's top-ranked chunks were on-topic but did not answer the question. Typical examples are a chunk that says Apple "manufactures" products without saying where, the general Additional Amounts rule instead of its exceptions, and the case history of the State Aid ruling instead of the amount paid.

Table 3: Gold chunk rank before (bi-encoder) and after (cross-encoder) re-ranking.

Category	Query	Gold	Bi	CE	Baseline #1
Keyword	Where are Apple's products manufactured?	239	#4	#1	c11 "designs, *manufactures* and markets"
Exception	Exceptions to obligation to pay Additional Amounts?	310	#4	#1	c309 general rule
Negation	Can Apple redeem Notes partially if tax laws change?	318	#4	#1	c279 exhibit index
Negation	Does Apple pay additional amounts to US holders?	309	#7	#2	c15 business overview
Buried chunk	Taxes paid due to EU State Aid ruling?	215	#5	#2	c214 case history (no $)
Buried chunk	Internal control changes in Q4?	263	#8	#1	c253 auditor opinion
Distractor	Shares repurchased in fiscal 2025?	230	#8	#2	c140 Q4-only table

2.2 Answer Quality

Better retrieval translated into better answers in most cases (Table 4): correct answers rose from 1/7 to 5/7. The baseline rarely hallucinated. It mostly abstained, because the refusal instruction worked when the answering chunk was missing. In the exception query it omitted the essential clause, noting only that exceptions exist "below".

Two queries stayed wrong after re-ranking. For partial redemption, the gold chunk was present but the LLM misstated the rule, claiming partial redemption is allowed when the text says "in whole, but not in part". For US holders, the rule is only implied (Additional Amounts are paid to holders who are *not* US persons) and the LLM did not draw the inference. Re-ranking fixes what reaches the LLM, not how the LLM reasons over negation.

2.3 Latency

Re-ranking ten pairs adds about 13.6 ms, so retrieval goes from 9.1 ms to 22.1 ms (Table 5). Both are small next to LLM generation. Pipeline 2's higher generation time reflects longer answers: the baseline often returns a one-line refusal. Qwen3's thinking mode was disabled, so these generation times cover direct answers only.

Table 4: Answer 1 (baseline) vs. Answer 2 (re-ranked).

Query	Answer 1	Answer 2	A1	A2
Manufactured	"Context does not contain this."	Outsourcing partners in China, India, Japan, S. Korea, Taiwan, Vietnam	×	✓
Additional Amounts exceptions	"Subject to exceptions… not detailed"	Lists the "shall not apply" carve-outs (US trade/business, US connection)	×	✓
Partial redemption	"Context does not contain this."	Says partial redemption *is* allowed for some series	×	×
US holders	"Context does not contain this."	"Context does not contain this."	×	×
State Aid tax	"Context does not contain this."	$10.2B net charge ($15.8B to Ireland, $4.8B US credit)	×	✓
Q4 control changes	"Context does not contain this."	No changes in Q4 2025	×	✓
Shares repurchased	401,672 thousand	402 million	✓	✓
Correct			**1/7**	**5/7**

Table 5: Mean latency over 10 queries.

Stage	Pipeline 1	Pipeline 2
Query embedding + FAISS search (ms)	9.08	8.46
Cross-encoder, 10 pairs (ms)	–	13.62
Retrieval total (ms)	**9.08**	**22.07**
LLM generation (s)	0.77	1.57

3 Discussion

3.1 Why Re-Ranking Helps

The failures in Table 3 share one pattern: the bi-encoder ranks by topical overlap. A chunk about Apple's business, or about Additional Amounts in general, is close to the query in embedding space even if it does not contain the answer. Words that change meaning, such as "not", "except" or "in whole", barely move a pooled embedding. The cross-encoder sees the query and chunk together, so it can check whether the chunk actually answers the question.

3.2 Where Re-Ranking Does Not Help

The cross-encoder can only re-order what the bi-encoder retrieves (Table 6). A paraphrased question about the European Court of Justice ruling never retrieved the chunk containing the $10.2B charge, and a question about the departing executive never retrieved Item 9B. In the ECJ case re-ranking was slightly harmful: it dropped the case-history chunk, so the baseline's partial answer became an abstention. In one case the cross-encoder also demoted a gold chunk (#6 → #7).

Table 6: Queries where re-ranking did not help.

Type	Query	Gold	Bi → CE	Result
Recall failure	Financial impact of the ECJ decision?	215	>10 → >10	A1 partial (case history); A2 abstains
Recall failure	Which executive is departing?	264	>10 → >10	Both abstain
CE demotes gold	Which taxes are excluded from Additional Amounts?	313	#6 → #7	Same incomplete answer

3.3 Scaling

The cross-encoder is not run over the whole corpus because its cost grows linearly with corpus size on every query. Its scores depend on the query, so nothing can be pre-computed. Bi-encoder chunk vectors are computed once offline, and search over them is fast and can be sub-linear with approximate indexes. Table 7 extrapolates my measured cost of ≈1.36 ms per pair. Even this small document would take ∼0.5 s per query, and a 1M-chunk collection would take tens of minutes. The two-stage design uses the cheap bi-encoder for recall and spends the expensive cross-encoder only on k candidates.

Table 7: Per-query cost of full-corpus cross-encoding (extrapolated from 1.36 ms/pair).

Corpus size	Bi-encoder search	Cross-encoder on all chunks
376 chunks (this document)	∼9 ms	∼0.5 s
100k chunks	∼10 ms (ANN)	∼2.3 min
1M chunks	∼10 ms (ANN)	∼23 min

3.4 Limitations

- Small evaluation set: 7 failure queries and 3 limitation queries on one document, so they show failure modes rather than average accuracy.
- Correctness was judged manually against a single gold chunk per query.
- Financial tables are flattened by text extraction, so table-lookup questions were avoided.